Roll No. .....

Total No. of Questions : 5] [Total No. of Printed Pages : 04

ZE-199

B.Tech. (New Scheme) (BE)

Examination, 2026

(Fifth Semester)

CONTROL SYSTEM

EL-503

Time : 3 Hours] **[Maximum Marks : 60**

Note : Attempt all questions. Each question carries equal marks.

1. (a) Explain the following terms with reference to block diagram representation :

   (i) Block diagram

   (ii) Output

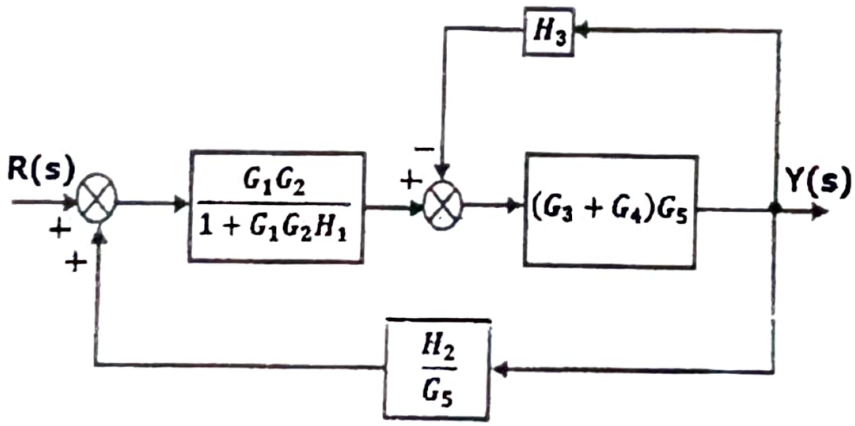
   (iii) Summing point

(iv) Take off point

(v) Forward path

(vi) Feedback path

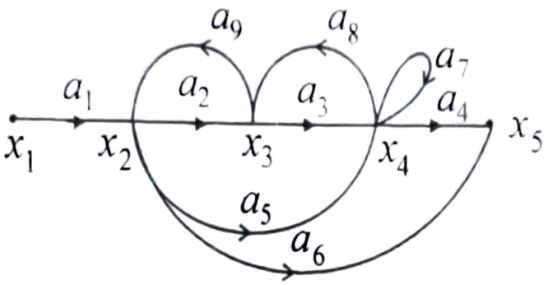
(b) The block diagram of a closed-loop system is shown in figure. Using the block diagram reduction technique determine the closed-loop transfer function C(s)/R(s).



Or

(a) Draw the block diagram of open loop and close-loop control system and campare them. Give them advantages and disadvantages of each.

(b) Find the overall C(s)/R(s) for the signal flow graph shown in fig. :



2. (a) Compare the stead state error characteristics of type 0, type 1 and type 2 systems.

(b) Determine the damping ratio and undamped natural frequency of oscillatory roots and %Mp for a unit step input given that :

$$G(s) = \frac{k}{s(s+2)(s+8)}$$

Or

(a) What will be the nature of response of a second order system with different types of damping ?

(b) For a system whose $G(s) = 10/s(s + 1)(s + 2)$. Find the steady state error when it is subjected to the input, $r(t) = 1 + 2t + 1.5t^2$.

3. (a) What is Root Locus ? What are breakaway and Breakin point ?

(b) For a control system, the gain crossover frequency is 5 rad/sec and the phase crossover frequency is 10 rad/sec. Find the gain and phase margin if the phase at 5 rad/sec is –140 degree.

Or

(a) Define absolute stability in the context of control systems. Explain how the pole locations of a sytem influence absolute stability.

(b) Sketch the root locus for the unity feedback system whose open-loop transfer function is :

$$G(s)H(s) = K/s(s + 4)(s^2 + 4s + 20)$$

4. (a) What is bode plot and why is it used in control system ? Also define gain crossover Frequency and phase crossover frequency.

(b) Sketch the bode plot of the given open-loop transfer function and from the plot determine the phase margin and gain margin.

$$G(s)\ H(s) = 50/(s + 1)\ (s + 2)$$

Or

(a) Establish correlation between time response and frequency response parameters.

(b) Sketch the bode plot for the following open-loop control system :

$$G(s)H(s) = \frac{k(1+0.2s)(1+0.025s)}{s^3(1+0.001s)(1+0.005s)}$$

5. (a) Define Observability. What is the need for observability test ?

(b) Convert the following transfer function into state space form :

$$\frac{y(s)}{r(s)} = \frac{s^2 + 2s + 4}{s^3 + 3s^2 + 6s + 10}$$

Or

Write short notes on the following

(i) Controllability

(ii) Transfer function decomposition

(iii) Solution of state equation

(iv) All pass and minimum phase system.

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ZE-199-A

B. Tech. ATKT Examination, 2026

(Fifth Semester)

ELECTRONICS AND COMMUNICATION ENGINEERING

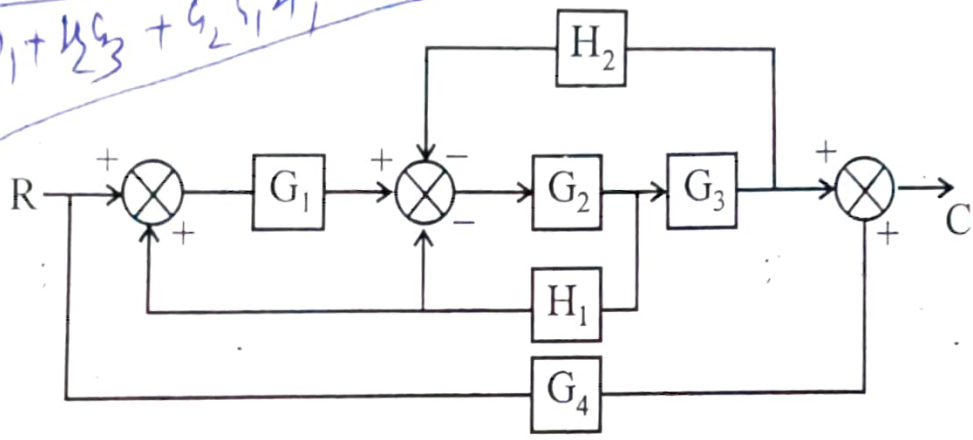
EL-503

Control System

Time : 3 Hours] **[Maximum Marks : 60**

Note : Attempt any *Five* questions. All questions carry equal marks.

1. (a) Obtain the transfer function c/r for the block diagram shown.



(b) Explain dc servo motor & pid controllers.

2. (a) What is open loop and closed loop control system ? Give their advantage, disadvantage and application.

(b) For a system having G(S) = 2S/S(S + 10) with unit negative feedback determine :

(i) Natural frequency

(ii) Damping ratio

(iii) Peak time

3. (a) For a unity feedback system having $G(S) = 3S(S + 6)/S(S + 1)(S + 7)$) determine :

(i) Type of system

(ii) All error cofficient

(iii) Steady state error for unit ramp input

(b) Explain time response of first, second and higher order system to various test signals.

4. (a) Explain hurwitz criterion & nyquist criterion.

(b) Determine the range of K so that the system having the following characteristic equation will be stable :

$$S(S^2 + 2S + 3)(S + 2) + K = 0$$

5. (a) Discuss the concept of relative stability & absolute stability.

(b) Sketch the root locus os the system when open loop transfer function is given by :

$$G(S)H(S) = K(S + 2)/S^2(S + 3)$$

6. (a) A system is described by differential equation :

$$\frac{d^3y}{dt^2}+3\frac{d^2y}{dt^2}+17\frac{dy}{dt}+5y=10u(t)$$

where y is the output & u is the input to the system Determine the state space representation of the system.

(b) Obtain the state transition matrix if :

$$A=\begin{bmatrix}0 & 1\\ -6 & 5\end{bmatrix}$$

7. (a) What is state space analysis ? Explain controllability and operatibility.

(b) Convert the following system matrix to conical form and hence calculate the transition matrix E^{AT} :

$$A=\begin{bmatrix}4 & 1 & -2\\ 1 & 0 & 2\\ 1 & -1 & 3\end{bmatrix}$$

8. Write short notes on any *three* :

(i) AC synchronous tachometer

(ii) Z-transform

(iii) Potentiometer

(iv) Phase lag lead compensation

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